

THALUWA BAZAR — ADMIN PANEL SPECIFICATION

Enterprise Platform Operations, KYC Verification, Order Moderation & Financial Settlement Architecture
Mission-Critical Governance Engine for Assam's Hyperlocal Rural-Urban Marketplace Startup (Darrang Pilot)

1. Platform Governance Vision & Admin Panel Scope

Thaluwa Bazar connects rural Self-Help Groups (SHGs), organic farmers, handloom artisans, and fishermen across Assam directly with urban buyers. While buyer and seller mobile clients prioritize frictionless ordering, the **Admin & Moderation Infrastructure** acts as the central authority—ensuring producer authenticity, financial clarity, order dispute resolution, and regulatory compliance.

**Core Administrative Pillars**

- Producer Authenticity:** Rapid audit and verified badge awards for SHG collectives, weavers, and organic growers.
- Weekly Haat Hub Control:** Schedule configuration for bi-weekly rural haats (Mangaldai, Sipajhar, Kharupetia).
- Order Dispute Moderation:** Real-time oversight on state transitions with administrative overrides.
- Financial Settlement:** Reconciling Cash-on-Delivery (COD) with direct UPI/bank payouts.

**Administrative Role Hierarchy**

- Super Administrator:** System configuration, batch financial disbursal, and platform analytics.
- District Moderator:** KYC verification of local SHGs, listing quality control, and dispute resolution.
- Haat Coordinator:** Physical market attendance, stall check-in, and COD cash aggregation.
- Automated Guard Engine:** Background processes computing GMV, stock levels, and push triggers.

2. Technology Stack & Security Architecture

Layer	Technologies & Libraries	Administrative Purpose
Mobile Ops Client	React Native, Expo SDK 52, Expo Router (<code>src/app/admin/index.tsx</code>), Lucide Icons, Haptics	Field-ready mobile dashboard for district moderators visiting rural SHG clusters and weekly haats.
API & Logic Engine	Node.js 20+, Express, TypeScript, <code>authMiddleware.ts</code> , <code>adminController.ts</code>	Executes multi-table aggregations, verification transitions, and automated push notifications.
Database & RLS	Supabase PostgreSQL with PostGIS, Service Role Key (<code>supabaseAdmin</code>)	Privileged data layer bypassing tenant RLS for cross-district analytics, KYC updates, and audit logs.
Real-Time Alerts	Expo Push Notification Service + <code>public.notifications</code> (Bilingual: Assamese/English)	Instant push alerts to seller mobile devices upon KYC verification or order intervention.

 **Startup Security Directive:** Field moderators cannot perform administrative mutations via client-side Supabase keys. All admin operations (approvals, rejections, fee adjustments) must route through the authenticated Node.js backend using validated JWT tokens and role verification (`users.role IN ('admin', 'moderator')`).

3. Live Metrics & Operational Analytics Aggregation

The Admin Dashboard renders a real-time **2x2 Executive KPI Grid** calculated via parallel non-blocking queries on the Node.js backend (`Promise.all`) against Supabase:

METRIC 1

**Total Users**

`SELECT count(*) FROM users .`
Monitors platform user growth.

METRIC 2

**Verified vs Pending**

Breaks down `seller_profiles` into active vs unreviewed queue.

METRIC 3

**Total Orders**

Tracks total transaction volume and active fulfillment states.

METRIC 4

₹ Platform GMV

Sum of `orders.total` for completed sales formatted in ₹ (e.g. ₹2,45,680).

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Section 4, 5 & 6: Administrative Data Model, KYC Lifecycle & Financial Settlement

4. Administrative Database Architecture (PostgreSQL / Supabase)

The administrative data model extends the core transactional schema with verification states, audit logs, and payout records:

Table Name	Administrative Fields & Constraints	Governance Function
public.users	id, phone, role ('buyer' 'seller' 'moderator' 'admin'), verified, expo_push_token, language	Enforces RBAC access guards and holds push notification tokens for admin broadcasts.
public.seller_profiles	id, user_id, business_name, shg_code, seller_type, verification_status, bank_details, verified_at	Maintains KYC lifecycle (pending → under_review → verified rejected).
public.orders	id, buyer_id, seller_id, total, status, dispute_status, override_by, commission_fee, created_at	Global ledger enabling administrative overrides (e.g., cancelling stuck orders, issuing refunds).
public.seller_payouts	id, seller_id, amount, status ('pending' 'processed' 'failed'), utr_ref, disbursed_at	Tracks weekly bank/UPI disbursements to rural SHG cooperative accounts.
public.haat_markets	id, name_en, name_as, district, market_day, geom, is_active, coordinator_id	Controls schedule and location coordinates for weekly rural market aggregation points.
public.audit_logs	id, actor_id, action, target_table, target_id, changes_json, ip_address, created_at	Immutable audit log capturing all moderator interventions and KYC decisions.

5. Critical Workflow: Seller KYC & SHG Verification Pipeline

To protect buyers from counterfeit goods and preserve Assamese craft authenticity, all sellers undergo an administrative verification pipeline before receiving the *Verified Seller Badge* (প্ৰমাণিত বিক্ৰেতা):

KYC Verification State Machine & Automated Notification Lifecycle:
[Seller Submits SHG Code/ID] → pending → [Moderator Inspects Registry] → under_review → [Admin Decision] → verified OR rejected

STEP 1
Queue Ingestion
App queries GET /api/v1/admin/pending-sellers . Fetches unverified seller profiles joined with user details.

STEP 2
Document Audit
Moderator validates SHG Code (e.g. AS-MRG-SHG-2024) against State Rural Livelihood Mission records.

STEP 3
Atomic DB Update
Calls POST /api/v1/admin/sellers/:id/status with status: 'verified' . Sets users.verified = true .

STEP 4
Bilingual Push Alert
Dispatches Expo Push + in-app notification in Assamese: "বিক্ৰেতা প্ৰফাইল প্ৰমাণিত হৈছে!"

6. Financial Settlement & Commission Reconciliation Engine

Thaluwa Bazar operates on a low-commission rural enablement model (0% commission for direct SHG farmers; 3–5% for commercial aggregators). The Admin panel manages the settlement lifecycle:

Cash-on-Delivery (COD) Reconciliation
When orders are delivered at haats or doorsteps, COD collections are logged by Haat Coordinators. The system offsets cash collected against the seller's digital balance before weekly payouts.

Batch NEFT/UPI Disbursal
Every Monday, the backend generates an automated payout ledger for all COMPLETED orders. Admins trigger batch bank transfers with UTR logging into seller_payouts .

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Section 7, 8 & 9: API Reference, Mobile Interface & Execution Roadmap

7. RESTful Admin API Specifications

Endpoint & Method	Auth Level	Request Body / Params	Response Structure
GET /api/v1/admin/stats	Admin/Mod	None	{ totalUsers, totalSellers, verifiedSellers, pendingSellers, totalOrders, totalGmv, formattedGmv }
GET /api/v1/admin/pending-sellers	Admin/Mod	Query: limit, district	{ success: true, count: N, pending_sellers: [{ id, business_name, shg_code, users: {...} }] }
POST /api/v1/admin/sellers/:id/status	Admin/Mod	{ status: 'verified' 'rejected', remarks: '...' }	{ success: true, seller_profile: {...} } + triggers push alert to seller device.
GET /api/v1/admin/orders	Admin/Mod	Query: limit=100, status, district	{ success: true, count: N, orders: [{ id, order_number, buyer_name, total, status, items: [...] }] }
POST /api/v1/admin/orders/:id/override	Admin Only	{ new_status, reason, refund_amount }	{ success: true, updated_order: {...}, audit_id }
GET /api/v1/admin/haats/schedule	Public/Mod	Query: district, active_only	{ success: true, haats: [{ id, name_as, name_en, district, market_day, coordinator }] }

8. Mobile Admin Interface Architecture (Expo Router)

 **Native Screen Hierarchy**

- src/app/admin/index.tsx : Main Dashboard (2x2 KPI grid, pending seller swipe-to-approve, top revenue leaders, recent order stream).
- src/app/admin/seller-detail.tsx : Deep-dive view with SHG registration photo preview, UPI ID verification, and rejection modal.
- src/app/admin/orders.tsx : Live order management filterable by status (REQUESTED, ACCEPTED, READY, COMPLETED).

 **UX & Native Performance Highlights**

- Haptic Feedback:** expo-haptics triggers distinct vibration signatures for successful KYC approvals vs. rejections.
- Pull-to-Refresh:** RefreshControl enables instant pull down to synchronize live stats and pending queues.
- Bilingual Fallback:** Assamese & English string translations via LanguageContext for localized field operations.

9. Phased Admin Rollout & Engineering Roadmap

Phase	Target Deliverable	Key Architectural Components	Status
Phase 1	Mobile Admin Dashboard & Live Stats	React Native 2x2 grid, Node.js /admin/stats endpoint, Supabase count aggregations.	COMPLETED
Phase 2	KYC Verification & Push Alerts	Pending seller queue, approval mutation, bilingual Expo Push Notification dispatch.	COMPLETED
Phase 3	Order Moderation & Haat Schedules	Global orders ledger, status intervention overrides, weekly market hub activation.	IN PROGRESS
Phase 4	Automated SHG Batch Payouts	RazorpayX / Cashfree Payouts API integration for automated bank disburseals.	PLANNED
Phase 5	Web Ops Portal for District Officials	Vite/React Desktop dashboard for Darrang district agricultural officers and co-op supervisors.	PLANNED

🚀 **Production Deployment Note:** When deploying backend to cloud production (Render / Railway / AWS), ensure SUPABASE_SERVICE_ROLE_KEY and EXPO_ACCESS_TOKEN are secured in environment secret managers. Restrict admin routes with strict IP rate limiting (100 reqs/min per IP).